OUARDIA BELAID

Recent Master's Graduate | Networks, Mobility & Embedded Systems
Algiers, Algeria

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PROFILE

Recent Master's graduate in Networks, Mobility, and Embedded Systems, with a strong interest in networks, IoT, cybersecurity, and artificial intelligence. Through academic projects and my Master's thesis on deep learning-based intrusion detection in IoT networks, I have gained hands-on experience in embedded systems, network simulation and security, AI, as well as web and mobile development. Eager to apply my technical knowledge and continue developing my skills, I am seeking a first professional opportunity, either through an entry-level position or an internship.

EDUCATION

❖ **Mouloud Mammeri University – Tizi Ouzou** ..... **2024 - 2026**
Master's degree in Networks, Mobility, and Embedded Systems

- Networks & Security, Networks & QoS, Mobile Networks, Wireless Sensor Networks
- Operating Systems, Embedded Systems, Assembly Programming, VHDL, Real Time Operating System
- Parallel Architecture & Distributed Systems, Parallel Programming
- AI Fundamentals, Image Processing & Computer Vision

❖ **Mouloud Mammeri University – Tizi Ouzou** ..... **2022 - 2024**
Bachelor's degree in computer science

- Web Development, Mobile Development, Artificial Intelligence, Linear Programming
- Computer Architecture, Networks, Software Engineering, Databases
- Operating Systems, Security, Compilation

❖ **National Higher School of Computer Science (ESI Oued Smar) – Algiers** ..... **2019 - 2022**
Integrated Preparatory Classes

❖ **Technicum Oudni Aomar High School – Draâ El Mizan** **2019**
Baccalaureate in Mathematics

ACADEMIC PROJECTS

❖ **Master Thesis : Intrusion detection in IoT systems using Deep Learning**..... **Mars - September 2026**

- Built and evaluated three Deep Learning architectures: 1D-CNN, GRU, and hybrid 1D-CNN+GRU for multiclass intrusion detection.
- Conducted comparative experiments on CICIoT2023 and ToN-IoT datasets, investigating class imbalance handling and model generalization.
- Evaluated detection performance using classification metrics, confusion matrices, and computational cost analysis.

Tools : Python, PyTorch, Pandas, Scikit-learn,matplotlib, Seaborn, Kaggle.

❖ **Remote-Controlled Rover with BLE** .....**February 2026**

- Developed an ESP32-based mobile rover controlled via a smartphone over Bluetooth Low Energy (BLE), using FreeRTOS for real-time task management.

Tools: ESP32, C, FreeRTOS, ESP-IDF, BLE, React Native, Expo.

❖ **Weather Forecast Mobile Application** ..... **May 2025**

- Developed a mobile application featuring real-time weather data, hourly and five-day forecasts, weather and time-based UI and lifestyle tips, localisation permissions handling for location based information and search history.

Tools: React Native, Typescript, Expo, OpenWeatherMap API.

❖ **STM32 Projects (Keil uVision)** ..... **February 2025 - April 2025**

- Implemented a multiplexed seven-segment stopwatch and an ADC-based temperature monitoring system using low-level C programming, GPIO, and interrupts on the STM32F407.

Tools: STM32F407, C, Keil µVision, ADC, GPIO, SysTick, temperature sensor.

❖ **Pottery E-Commerce Website - Bachelor Degree Project** ..... **Mars 2024 — May 2024**

- Developed a web solution for showcasing and selling handcrafted pottery, with product, order, and customer management features.

Tools: Django, Bootstrap, SQLite, HTML, CSS, JavaScript.

SKILLS

- ❖ **Networks & Cybersecurity:** Network protocols – Virtualization: VirtualBox, VMWare – Security: Wireshark, Nmap, Nessus – Network configuration
- ❖ **Programming Languages:** C, Python, Java, Assembly, Matlab.
- ❖ **Embedded Systems & IoT:** Low-level development on STM32 and ESP32 (C) – GPIO programming, Timers, communication protocols (I2C, SPI) – VHDL – IoT applications – RTOS(FreeRTOS).
- ❖ **Operating Systems:** Linux (Ubuntu, Xubuntu, Kali linux), Windows, WSL.
- ❖ **Tools:** VSCode, Keil, Quartus II, Isis Proteus, OMNET++, Packet Tracer, Git & GitHub, Miro, ClickUp.
- ❖ **Web Development:** HTML, CSS, Django.
- ❖ **Mobile Development:** React Native & Expo, Java (Android).
- ❖ **Databases:** PostgreSQL, SQLite.
- ❖ **Data, IA & Image Processing:** Python (NumPy, Matplotlib, Scikit-learn, Pandas,Pytorch) - Machine Learning & Deep Learning.
- ❖ **Soft skills:** Problem solving, Team work, Attention to detail.

COMMUNITY & VOLUNTEERING

❖ CSI CTF Participant ..... Mars 2024 & June 2026

❖ WTM Algiers Manager ..... 2022 — 2023

❖ Events Co-Manager, GDG Algiers & WTM Algiers ..... 2021 — 2022

❖ Women TechMakers Ambassador ..... 2021 — 2025

❖ Active Member, Logistics Department – GDG Algiers & WTM Algiers ..... 2020 — 2021

CERTIFICATIONS

❖ Computer Networks CRA Training Program - Huawei ICT Academy ..... Nov 2025

❖ Fundamentals of Deep Learning – Nvidia ..... June 2025

❖ Google IT Support Certificate : IT Support Fundamentals - Coursera / Google ..... April 2022

LANGUAGES

English - French - Arabic - Kabyle